

Paper Practice 3.3**Linearity and Continuity**

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. Simplify before you decide – an equation can look nonlinear and simplify to a line.

- 1.** Determine whether each function is linear or nonlinear. Simplify first. (Simplify First)

a. $y = 4x^2 - (2x)^2 + 3x - 5$ ANSWER:
Linear. $(2x)^2 = 4x^2$, so $4x^2 - 4x^2 = 0$, leaving $y = 3x - 5$.

b. $y = 3x^3 - x^2 + 3x + 6$ ANSWER:
Nonlinear – the x^3 term does not cancel.

c. $4x - (2y)^2 = 3$ ANSWER: Nonlinear.
 $(2y)^2 = 4y^2$, and y is squared.

- 2.** Determine whether each function is linear or nonlinear. (Linear or Not)

a. $2x + 5y = 10$

b. $y = x^2 - 4$

c. $y = 7$

- 3.** Simplify, then decide whether the function is linear or nonlinear. (Simplify First)

a. $y = 6x^2 - (3x)^2 + 3x^2 + 4x + 1$
ANSWER: Linear. $6x^2 - 9x^2 + 3x^2 = 0$,
leaving $y = 4x + 1$.

- 4.** Write each linear equation in standard form $ax + by = c$ with integer coefficients. (Standard Form)

a. $y = 3x - 8$

b. $y = -\frac{1}{2}x + 4$

- 5.** Determine whether each situation is discrete or continuous. Explain. (Discrete or Continuous)

a. the number of tickets sold for a concert
ANSWER: Discrete – you cannot sell part of a ticket.

b. the temperature of a room over one hour ANSWER: Continuous –
temperature takes every value in between.

c. the number of students riding a bus
ANSWER: Discrete – students are counted in whole numbers.

- 6.** The Bargain Book Barn sells novels on a sliding scale: the more you buy, the cheaper each one is. Let $f(x)$ be the price for x books. Is $f(x)$ discrete or continuous? Explain. (Discrete or Continuous)

- 7.** A function has domain $\{0, 1, 2, 3\}$ and range $\{2, 5, 8, 11\}$. Is it discrete or continuous? Explain.
(Discrete or Continuous) **ANSWER:** Discrete – the domain is a finite set of separated points.

- 8.** The table shows a function. Is it linear? Justify your answer using the rate of change. (Rate of Change)

a. x: 1, 2, 3, 4 y: 3, 6, 9, 12

- 9.** The table shows a function. Is it linear? Justify your answer. (Rate of Change)

a. x: 1, 2, 3, 4 y: 2, 4, 8, 16

ANSWER: Nonlinear – the differences are 2, 4, 8. The rate of change is not constant.

- 10.** Give the domain and the range of the discrete function $\{(-2, 1), (0, 3), (2, 5)\}$. (Domain & Range)

- 11.** Error Analysis – Gus says $y = 4x^2 - (2x)^2 + 3x - 5$ is nonlinear because it contains x^2 . Identify his mistake and give the correct answer. (Error Analysis) **ANSWER:** He never simplified. $(2x)^2 = 4x^2$, so the x^2 terms cancel and $y = 3x - 5$, which is linear.

- 12.** Explain the difference between a discrete and a continuous function, then give a real-world example of each that is not used above. (Explain)